Ref.: Ms. No. FISH13888

Development of a Simple R Package (aLBI) for the Estimation of Stock Status from the Length Frequency Data

Fisheries Research

Dear Dr. Sarker,
I can now inform you that the Editorial Board has evaluated the manuscript FISH13888: Development of a Simple R Package (aLBI) for the Estimation of Stock Status from the Length Frequency Data.

The Editor has advised that the manuscript will be reconsidered for publication after major revision.

The comments below should be taken into account when revising the manuscript. Along with your revised manuscript, you will need to supply Revision notes in which you list all the changes you have made to the manuscript, and in which you detail your responses to all the comments passed by the reviewer(s) and the Editor. Should you disagree with any comment(s), please explain why.

To submit a revision, please visit https://www.editorialmanager.com/fisheries/ and log in as an Author. You will see a menu item called Submission Needing Revision. The revised manuscript and covering letter can be submitted there.

When submitting your revised manuscript, please ensure that you upload the source files (e.g. Word). Uploading a PDF file at this stage will create delays should your manuscript be finally accepted for publication. If your revised submission does not include the source files, we will contact you to request them.

You are kindly requested to submit your revised manuscript within 4 weeks. If your revision is received after that deadline, it may be treated as a new submission.

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Kind regards,

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Note: While submitting the revised manuscript, please double check the author names provided in the submission so that authorship related changes are made in the revision stage. If your manuscript is accepted, any authorship change will involve approval from co-authors and respective editor handling the submission and this may cause a significant delay in publishing your manuscript.

Important note: If a reviewer has provided a review or other materials as attachments, those items will not be in this letter. Please ensure therefore that you log on to the journal site and check if any attachments have been provided.

Reviewers' comments:

Reviewer's Responses to Questions

Note: In order to effectively convey your recommendations for improvement to the author(s), and help editors make well-informed and efficient decisions, we ask you to answer the following specific questions about the manuscript and provide additional suggestions where appropriate.

1. Are the objectives and the rationale of the study clearly stated?

Please provide suggestions to the author(s) on how to improve the clarity of the objectives and rationale of the study. Please number each suggestion so that author(s) can more easily respond.

Reviewer #1: The objective of the study to develop and introduce an R package that contains length-based indicators for fisheries analysis is clearly stated.

Reviewer #2: yes

2. If applicable, is the application/theory/method/study reported in sufficient detail to allow for its replicability and/or reproducibility?

Please provide suggestions to the author(s) on how to improve the replicability/reproducibility of their study. Please number each suggestion so that the author(s) can more easily respond.

Reviewer #1: Mark as appropriate with an X:
Yes [X] No [] N/A []
Provide further comments here:
Reviewer #2: Mark as appropriate with an X:
Yes [] No [] N/A []
Provide further comments here:
not applicable

3. If applicable, are statistical analyses, controls, sampling mechanism, and statistical reporting (e.g., P-values, CIs, effect sizes) appropriate and well described?

Please clearly indicate if the manuscript requires additional peer review by a statistician. Kindly provide suggestions to the author(s) on how to improve the statistical analyses, controls, sampling mechanism, or statistical reporting. Please number each suggestion so that the author(s) can more easily respond.

Reviewer #1: Mark as appropriate with an X:
Yes [] No [X] N/A []
Provide further comments here:

The manuscript does not include any details on the bootstrapping or aggregation of length data to length bins. Different bin sizes for the length frequency aggregation will affect the estimation of maximum length and other length indicators as well as the bootstrapping. The equations are incomplete, P_L in equation 4-6 is not defined.

Reviewer #2: Mark as appropriate with an X:
Yes [] No [] N/A []
Provide further comments here:
not applicable

4. Could the manuscript benefit from additional tables or figures, or from improving or removing (some of the) existing ones?

Please provide specific suggestions for improvements, removals, or additions of figures or tables. Please number each suggestion so that author(s) can more easily respond.

Reviewer #1: The manuscript could benefit from removing the details about the R package development and the part of the manuscript that describes the details of the package, which fits better into a package vignette or user manual than a scientific article. Instead, the manuscript could evaluate the benefits of the added functionality to LBIs, e.g. bootstrapping, and demonstrate the uncertainty in indicators (stock status proxies) for various data scenarios.

Reviewer #2: may delete Figure 1

5. If applicable, are the interpretation of results and study conclusions supported by the data?

Please provide suggestions (if needed) to the author(s) on how to improve, tone down, or expand the study interpretations/conclusions. Please number each suggestion so that the author(s) can more easily respond.

Reviewer #1: Mark as appropriate with an X:

Yes [X] No [] N/A []
Provide further comments here:

However, uncertainty in final stock status proxies are not provided. It would be interesting and novel if the authors develop an approach to propagate the uncertainty of L_opt, L_mat, etc. to P_mat, P_opt, and P_mega.

Reviewer #2: Mark as appropriate with an X:

Yes [Y] No [] N/A []
Provide further comments here:

6. Have the authors clearly emphasized the strengths of their study/theory/methods/argument?

Please provide suggestions to the author(s) on how to better emphasize the strengths of their study. Please number each suggestion so that the author(s) can more easily respond.

Reviewer #1: No, I think the value of the bootstrapping could be emphasized and shifting the focus from package development and user manual to uncertainty estimation and testing, evaluation of indicators could strengthen this manuscript.

Reviewer #2: yes

7. Have the authors clearly stated the limitations of their study/theory/methods/argument?

Please list the limitations that the author(s) need to add or emphasize. Please number each limitation so that author(s) can more easily respond.

Reviewer #1: No, LBIs come with a long list of assumptions, such as equilibrium conditions, which are not mentioned by the authors. Given the usefulness of these methods in data-limited scenarios, it is crucial to note and communicate the assumptions of these methods.

Reviewer #2: yes

8. Does the manuscript structure, flow or writing need improving (e.g., the addition of subheadings, shortening of text, reorganization of sections, or moving details from one section to another)?

Please provide suggestions to the author(s) on how to improve the manuscript structure and flow. Please number each suggestion so that author(s) can more easily respond.

Reviewer #1: Yes, the manuscript should be carefully reviewed and flow and structure improved after excluding package development description. For example, specific length-based indicator values are given in the abstract, but the species, stock they correspond to is not mentioned.

Reviewer #2: may write more about the package, less about the application.

9. Could the manuscript benefit from language editing?

Reviewer #1: Yes
Reviewer #2: No

Thank you to both reviewers for their comments.

While they reached different conclusions about the readiness of the paper for application, I believe it is worth offering the authors a chance to respond to the major comments from reviewer #1.

There is a discrepancy between how the reviewers think the article space should be used, with reviewer #1 wanting less information about package development and reviewer #2 wanting more about the package.

A way to bridge these comments is to focus on the package capabilities and lose the developmental details as suggested by reviewer #1. I see the role of the case study to demonstrate how the package works. Because of this demonstration, reviewer #1 was able to note some very important details, such as assumptions and uncertainty estimation, that really need to be included. I agree with much of review #1 comments on how to improve the presentation on how the package works, and less on how it became a package.

Please respond to each reviewers comments in turn so the reviewers can track your response to their suggestions.

I look forward to receiving an updated manuscript.

Reviewer #1: The manuscript contains some interesting aspects, such as the bootstrapping of length frequency data to estimate uncertainty around length-based indicators and the estimation of stock status proxies for a data-limited stock. However, in its current form, the manuscript focuses too much on the R package development and introduction which would be more fitting for an application type article, a vignette or user manual and lacks scientific novelty. The manuscript could benefit from focusing less on the package development and instead focusing more on demonstrating the advantages of having these functions in a package and linked to bootstrapping, such as developing an approach to propagate uncertainty to the length-based stock status proxies (P_mat, P_opt, P_omega) and/or testing of the LBIs and associated uncertainty for different scenarios, such as different recruitment assumptions or data availability.

Reviewer #2: This is a simple and nice R Package (aLBI) for the estimation of stock status from the length frequency data, which is useful for the data-limited fisheries, I would recommend publication after some revisions.

I would suggest the authors to write more on the package, less on the application on the case study. For example, Figure 1 may be deleted. The authors may read the paper of TropFishR below as a reference.

Mildenberger, T. K., Taylor, M. H., & Wolff, M. (2017). *"TropFishR: An R package for 577 fisheries analysis with length-frequency data."* Methods in Ecology and Evolution, 578 8(11), 1520-1527.